The Illusion of the Given

Plain English Exposition, Multi-Dimensional Worldview, and Parallel Buddhist Traditions

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PART I: THE CORE TEXT IN SIMPLE LAYMAN'S ENGLISH

A Full, Paragraph-by-Paragraph Rendering

Introduction: The Puzzle of Agreement

Whenever different people look at the world and report the same features, our minds instinctively make a big assumption: we assume that this agreement must come from a solid, objective object standing quietly behind what we see. If two humans look at a river, or if a human, a dog, and an ant walk across the same patch of dirt, ordinary common sense insists there must be an underlying physical thing—an objective bedrock—that each creature simply registers or filters through its own biological senses.

In classical Indian and Tibetan philosophy, this puzzle is brought into sharp focus by a famous thought experiment: a single bowl of liquid seen by different kinds of beings.

- To an ordinary human, it appears as refreshing, thirst-quenching water.
- To a hungry ghost (a being consumed by intense habits of craving and lack), it appears as disgusting, burning pus and blood.
- To a fish or aquatic creature, it appears as an ambient atmosphere to breathe and a home to swim in.
- To a god in a celestial realm, it appears as heavenly, ambrosial nectar.
- To an awakened noble being, it appears as the radiant, pure wisdom-energy of water (the female Buddha Māmakī).

The immediate reflex of ordinary realism is to explain this difference through a two-layer, dualistic picture. We naturally imagine that some neutral, uncarved raw stuff (whether described as chemical H_2O molecules, bare matter, or pure physical energy) sits in that exact co-

ordinate space, and that differing organisms merely view that pre-existing substance through their own specialized sensory glasses.

However, both the Middle Way (Prāsaṅgika-Madhyamaka) and non-foundational Dzogchen (the Great Perfection) reject this assumption completely. When we examine it carefully, the idea of a shared thing that exists all by itself, beneath all the ways it appears, completely falls apart. Furthermore, this insight has strong echoes—though not proofs—in modern physics and the philosophy of science, specifically Carlo Rovelli's Relational Quantum Mechanics and Karen Barad's Agential Realism. These modern scientific views are used here only as helpful illustrations. If we were to turn around and treat relationships as the new "real stuff," we would simply be laying down another floor.

Agreement does not require a hidden bedrock. Orderly patterns do not require an ultimate floor. When conditions change, appearances change—not because different observers decode one untouched thing in different ways, but because the object seen, the observer looking, and the channel through which they interact are all designated together within dependent arising. The relationships between things are just as impossible to find as fixed things themselves.

1. The Trap of the Hidden Substance

When someone says that "different species see the same liquid differently," they slip in an unexamined noun: "the liquid." This claim quietly assumes that before anyone ever encounters it, a finished physical entity already sits there in quiet isolation, waiting to be inspected.

This is the classic retreat into the philosophy made famous by Immanuel Kant, known as transcendental idealism:

- **The Claim:** A hidden "thing-in-itself" exists completely beyond our minds, while living organisms simply stamp their own biological categories, spatial coordinates, and sensory habits onto it.
- **The Middle Way Critique:** If this hidden bedrock really exists prior to and independent of being experienced, what is it actually like? If it has definite properties (such as being refreshing water), it completely fails to explain why the hungry ghost directly encounters burning pus. If it has no properties at all, a featureless, property-free ghost cannot act as a real-world cause to trigger any sensory experience whatsoever.

Kant himself never claimed the thing-in-itself acts as a physical cause inside our everyday experience, though he called it the "non-sensible cause" of appearances—which is precisely where his critics found the logical flaw. And while Middle Way thinkers accept everyday cause and effect as practical convention, the core insight holds firm: a hidden substrate stripped of all relationships can explain nothing at all.

The "Wet-and-Moist" Counter-Argument

A thoughtful person might push back: "Why assume the hidden thing must be water or pus? Why couldn't it simply be 'wetness and moisture'—the basic physical quality common to both?"

This is a natural suggestion. Classical Buddhist psychology identifies wetness and moisture as the defining mark of the water element, common to whatever is fluid. But this suggestion faces a clear fork in the road:

1. **If wetness is claimed as something that exists on its own:** The argument fails. A generic, universal "wetness" is merely an abstract mental concept. In real life, wetness never floats in mid-air; it only ever shows up as this drinkable water, this caustic pus, or this fluid home for a fish. Furthermore, fluidity itself is a relationship: what acts as a fluid to a large biological creature with cell membranes does not function as a fluid to physical systems operating at subatomic scales or under different fundamental forces.
2. **If wetness is treated as an everyday practical reference point:** It is no longer a hidden thing-in-itself. It becomes a perfectly valid working label—a practical convention that lets human debaters compare and discuss different perspectives without pretending to point to an absolute, observer-independent floor.

As Nāgārjuna demonstrated, the real target of critical analysis is *svabhāva*: the belief in an intrinsic nature that exists all by itself, independent of anything else. Positing an uncarved "neutral stuff" beneath appearances is simply that old belief wearing a disguise.

2. Agreement Without a Hidden Substance: The Middle Way Account

If there is no inherently established object behind appearances, why do humans reliably agree that the liquid is water? Why doesn't our shared world collapse into chaos or private fantasy?

In Chapter 6 of his work *Entering the Middle Way*, the Indian master Candrakīrti provides the answer: agreement does not prove there is a hidden substrate beneath things; shared, reliable everyday coherence is the very definition of conventional truth.

A. Mind and Object Depend on Each Other

Candrakīrti spends substantial effort dismantling the "Mind-Only" view, which tries to solve the problem by getting rid of external matter and making an intrinsically real consciousness the sole foundation. Candrakīrti proves that consciousness and physical objects exist in strict mutual dependence:

- A perceiving mind cannot be identified without reference to an object that is perceived.
- An object cannot be identified without reference to the perceptual and conceptual framework that picks it out.

Neither mind nor matter comes first. Making the mind into an ultimate container is just as mistaken as making physical matter into the container.

B. Healthy Senses vs. Impairment Within Each Kind of Being

Candrakīrti distinguishes what the world accepts as true from what even the world recognizes as an error, based on whether a being's senses are healthy or impaired (for example, by an eye disease). Later thinkers called these correct and mistaken conventions. The classic examples are a person with jaundice who sees a white conch shell as yellow, or someone with an eye disease who sees floating hairs in the air.

Crucially, sensory health or impairment is defined strictly within a biological or karmic group:

- A human with jaundice sees the white conch as yellow. Because other humans with healthy eyes see it as white, the jaundiced person is conventionally mistaken by human standards.
- The hungry ghost, however, is not a "jaundiced human." Relative to the hungry ghost's own karmic conditioning, its senses are functioning normally. It is not suffering an optical illusion of water; for it, the liquid genuinely functions as burning pus.

Furthermore, this division into correct and mistaken is made purely from the viewpoint of everyday worldly life. Candrakīrti points out that awakened noble beings, having cleared away fundamental confusion, see compounded things as like reflections in a mirror: for them, these are mere conventions, not absolute truths.

C. A Working Reference Point

To avoid falling into nihilism (the mistaken belief that nothing exists), this analysis draws a clear distinction in line with Tsongkhapa's reading. The hungry ghost's perception of pus is treated as a real conventional perception shaped by karma, not as an optical mistake.

What Is Negated	What Is Conventionally Accepted
An inherently established common base: A mind-independent physical substance possessing its own standalone identity before anyone observes it.	**A mere working reference point:** A dependently designated convergence point where different karmic appearances meet and can be compared.

What prevents this working reference point from sliding back into a hidden "thing-in-itself"?

This working reference point is not a solid carrier of qualities that exists before its relationships. It has no qualities whatsoever apart from those that show up through a specific channel of contact. It is simply a practical label made from within an everyday frame—usually our human frame—which allows us to coordinate and compare different viewpoints. Its job is to coordinate perspectives, not to sit underneath them. In this respect it plays a role in the Middle Way similar to the one cross-perspective links play in relational quantum mechanics, rather than acting like a classical physical substance.

D. Where the Bowl Example Came From: Vasubandhu and Candrakīrti's Reply

The bowl example did not originate in the Middle Way. In his *Twenty Verses*, the philosopher Vasubandhu used the hungry ghosts to defend the Mind-Only school. Critics asked: if there are no external objects, why do our perceptions agree? Why do many people see the exact same thing at the same time and place? Vasubandhu answered that just as hungry ghosts with identical karma all see a river of pus together, shared karma ripening within consciousness accounts for shared perception. No river outside the mind is required.

That answers our core question by making consciousness the ultimate floor. Candrakīrti completely turns this example around. He takes the hungry ghost seeing pus, compares it to someone with an eye disease, and concludes that just as the external object does not exist on its own, neither does the mind. If the pus has no independent reality because it is not an external thing-in-itself, the mind seeing the pus has no independent reality either, because mind and object arise together. For the hungry ghost, the river of pus and the perception of pus stand or fall together. In everyday worldly terms, both mind and object are accepted; under ultimate analysis, neither can be found existing on its own. So the bowl does not prove that the world exists inside the mind. It proves that neither the world nor the mind is an ultimate floor. Shared karma explains why beings agree, but karma is simply one condition among many, not an absolute mental storehouse.

3. Other Observers

Agreement naturally requires other observers to agree with. This brings up three important questions: how we know other observers exist, whether this explanation is circular, and what role language plays.

A. Other Minds

Shared experience assumes there are other observers whose experiences could match mine. How do we know they exist? Indian Buddhist philosophers debated this extensively. Dharmakīrti argued that we infer other minds: other bodies move and speak with purpose just as mine does when I have an intention, so we infer their mental streams from their actions. Later, Ratnakīrti, arguing from the Mind-Only view, claimed that other minds can never finally be proven.

Both thinkers shared an assumption rejected here: that my own private mind is immediately given and certain, while other minds are merely inferred or doubtful. The Middle Way refuses this asymmetry. When analyzed, my own stream of consciousness is no more findable as an independent entity than yours is; both are designated in dependence on changing physical and mental parts, and both function perfectly in daily life. The certainty I feel about "my" experience and the doubt I feel about "yours" are just the old habit of reification turned inward, making the private self into the one unquestioned floor. In everyday life, the world accepts other minds just as naturally as it accepts tables. Other observers are not a conclu-

sion deduced from a private starting point; they are designated together with oneself, just as "long" is designated alongside "short."

B. Is Agreement Circular?

A critic might ask: faculties are considered healthy when they agree with other members of their group, and the group is defined by sharing those very faculties. The standards define the group, and the group defines the standards. Isn't that circular?

It is circular, and that circle is not a flaw. Circularity would be fatal only if agreement were required to rest on an outside foundation—an ultimate, non-participating judge. The Middle Way denies that any such judge exists. Our ways of knowing and the things known are established only in relation to one another. Because nothing is being built from an absolute ground up, there is no missing foundation for the circle to lack. Ludwig Wittgenstein reached a similar conclusion from another direction: what makes following a rule possible is agreement not in private opinions, but in a shared "form of life"; language requires agreement not just in dictionary definitions, but in everyday judgments. Agreement is not proof of an unmoving object behind the scenes; it is simply convention at work. This circle does not make errors impossible: inside our everyday framework, mistakes are easily corrected—a jaundiced eye is corrected by healthy companions, and a misreading is corrected by a second look. What the circle rules out is an impossible correction from outside every perspective.

C. Language

For human beings, the framework that keeps track of agreement is, above all, language. Words and labels are learned from others, corrected by others, and shared with others. A child learns "water" as a word that works: say the word, and a glass is brought. That is why humans agree so precisely not only that liquid is present, but that it is water, that it is cold, and that it is the same water as yesterday's. Labeling goes beyond spoken language: animals coordinate their worlds through perception and physical response without words. But human language carries our circle of agreement across vast distances and generations. It is also where the habit of creating solid entities is strongest: the noun "the bowl" tricks us into imagining an unchanging object standing behind every perspective. The very tool that coordinates our everyday worlds tempts us to believe it describes a single, hidden universe behind them.

4. The Experiential Shift: Non-Foundational Dzogchen

While the Middle Way uses sharp logical reasoning to dismantle rigid metaphysical beliefs, the Great Perfection (Dzogchen) describes how reality is directly met in living experience.

A. The Basis Is Not an Ontological Substrate

In the non-foundational Dzogchen teachings articulated by Longchenpa and Ju Mipham Rinpoche, the primordial basis is carefully distinguished from the karmic storehouse con-

sciousness (*ālaya*):

- The basis is described through three inseparable aspects: its essence, which is empty (primordial purity); its natural clarity, which is spontaneous presence appearing unobstructed; and compassion. We focus here on the first two.
- Primordial purity means that the basis has no solid, independent existence whatsoever. It is not an uncarved lump of cosmic stuff, a universal mind, or an ultimate starting point. Longchenpa explains that the basis is called a "ground" only from a conventional standpoint; in the actual nature of things, it is completely free from being a foundational floor.

B. Self-Display: Ordinary Mind vs. Pure Awareness

Appearances are described as "display" or "self-display." This word "display" must never be understood as the display of some hidden thing standing behind it. It simply means dependent arising itself: vivid, unobstructed, with nothing hidden beneath.

To prevent this from sliding into mentalism (the belief that everything is merely in the mind), Dzogchen draws a clear line between ordinary dualistic mind and pure awareness (*rigpa*):

- **The Grasping Subject Is Part of the Display:** Ordinary dualistic consciousness—the feeling of being an inside subject looking out at an external world—is itself an appearance. It arises symmetrically alongside the external object. Neither mind nor matter creates the other; both arise together within the display.
- **Awareness Is Not a Projector:** Pure awareness (*rigpa*) is not one half of this subject-object split, nor is it a subtle mental substance projecting outward forms. It is simply the immediate knowing of the basis: recognizing that whatever appears is fluid, dependent, and empty. The clarity of appearances belongs to the appearances themselves, not to a separate knower, and like the appearances, it is completely empty.
- **Display Is Not Causal Production:** The basis does not generate appearances the way a physical factory produces goods. Spontaneous presence is not an uncaused motor churning out conditioned things. Appearing is already complete as it is, owing its existence to nothing hidden, and it is empty from the very instant it appears.

C. Reflexive Awareness: A Deliberate Choice

Here the Buddhist traditions face a famous philosophical divide. In *Entering the Middle Way*, Candrakīrti rejected reflexive self-awareness (the idea that consciousness knows itself), using the classic metaphor that a sword cannot cut its own blade. Following Candrakīrti, Tsongkhapa rejected reflexive self-awareness even as a conventional description.

On this point, this analysis follows Ju Mipham Rinpoche:

- **Ultimately:** Mipham fully affirms the Middle Way critique. Under ultimate logical analysis, no self-knowing awareness that exists on its own can ever be found.

- **Conventionally:** Mipham defends reflexive awareness as a practical everyday term to explain memory and immediate experience without requiring an infinite chain of minds watching minds. This is not the truly existing self-awareness that Candrakīrti refuted; it is merely an everyday convention, and empty like every other convention.

In Dzogchen, the self-knowing of awareness is not a real subject knowing itself as an object. It is simply the natural absence of delusion: the immediate, unconstructed clarity of recognizing appearance as an empty display belonging to no one.

D. Mipham's Fourfold Epistemology and Pure Vision

Mipham organizes valid knowing into a fourfold framework:

1. **Impure Conventional Valid Knowing:** Operates within everyday karmic experience. Celestial beings belong here: their experience of liquid as nectar is an elevated karmic perception, not awakened vision.
2. **Pure Conventional Valid Knowing:** Belongs to awakened noble beings, where the liquid appears directly as its pure wisdom nature: Māmakī, the female Buddha of water.
3. **Conceptual Ultimate Valid Knowing:** The intellectual understanding of emptiness reached through rational deduction.
4. **Non-Conceptual Ultimate Valid Knowing:** The direct, living realization of the inseparable unity of primordial purity and spontaneous presence.

How does this fit with Candrakīrti's horizontal relativity? Mipham holds both together. Within ordinary, unawakened perception, validity is strictly relative to each kind of being: the hungry ghost's perception of pus is normal by hungry-ghost standards, and the human's water is normal by human standards. Neither has the final word over the other within everyday life. But between ordinary perception and awakened vision, there is a clear qualitative difference: pure vision is where how things appear matches how things actually are.

Does pure vision bring back a hidden foundation? No. Pure vision is itself completely empty—the inseparable union of appearance and emptiness. It is experience free from dualistic confusion and grasping; it is not a peek behind the curtain at a metaphysical bedrock. Māmakī herself has no inherent existence. Pure vision is accurate not because it finds an ultimate object, but because it meets appearances without the distorting habit that chops reality into separate subjects and solid, external things.

5. Echoes in Modern Physics: Rovelli and Barad

This dismantling of a hidden bedrock without falling into idealism has striking echoes in contemporary physics and philosophy. They are resonances, not proofs. Emptiness does not depend on physics, and scientific models are themselves practical designations—among the most reliable we possess within their fields.

Karen Barad's Agential Realism

- **No Things Before Relationships:** Traditional metaphysics presumes that independent objects exist first and then interact. In agential realism, the things interacting do not exist prior to their "intra-action"; they emerge through it.
- **The Agential Cut:** Boundaries and physical properties are enacted by specific measuring setups. What Barad calls the "agential cut" does not slice up a reality that was already there; it defines what counts as the observer and what counts as the observed in that specific situation.

There is no "liquid" standing there before the cut. The human drinker and the water emerge together within the human biological encounter; the suffering hungry ghost and the pus emerge together within the ghost's karmic encounter.

Carlo Rovelli's Relational Quantum Mechanics

In Rovelli's Relational Quantum Mechanics, physical variables do not have absolute values on their own; they take values only relative to another physical system during an interaction.

When critics asked why two observers agree when they compare notes if all values are relative, Emily Adlam and Carlo Rovelli introduced the Cross-Perspective Links postulate: if observer A measures a system and registers a value, and observer B later interacts with both the system and observer A, observer B's measurement will match observer A's record. Agreement between two observers does not require a hidden, unmeasured classical state to have existed beforehand. Agreement is established through physical interaction channels between the observers themselves. Shared reality is produced through relational coordination and cross-checking—a close counterpart to Candrakīrti's healthy senses confirming one another—not by anchoring to an absolute substrate.

> ***A Critical Caution:*** *Whatever their authors' own emphasis, neither quantum mechanics nor agential realism should be taken here as showing that "relations" are what really exists. Making the relationship into a new primary substance repeats the very foundationalism it sought to overcome. As Nāgārjuna showed, relationships and the things related arise together; neither comes first, and neither is an absolute floor.*

6. Different Channels of Contact

If an appearance requires specific senses, conditions, and interaction channels, what happens when a being encounters an environment with which it shares little or no contact?

The answer is neither "absolutely nothing" (which would make reality private and erase everyday cause and effect) nor a hidden classical object (which would bring back an absolute floor). Contact comes in degrees, and what holds true for any system depends entirely on which physical and cognitive channels are active.

A. Jakob von Uexküll's Tick

In biology, Jakob von Uexküll showed that an animal does not inhabit an abstract, neutral environment, but an *Umwelt*—a functional world carved out by its own sensory-effector loops:

- The common tick responds primarily to three signals: the smell of mammal sweat (butyric acid), the warmth of a living body, and hair.
- A pool of clear spring water has none of these signals, so it does not exist within the tick's perceptual world.

Yet the water is not "nothing" to the tick: a tick that falls into it is held by its surface and carried along by its current. The tick is uncoupled from the optical and perceptual channels of the water, but it remains fully coupled to its surface tension and fluid dynamics. This demonstrates why conventional reality is not trapped inside a private mind: cause and effect operate along specific relational channels whether an organism is consciously aware of them or not.

B. Different Channels in Physics

- **Dark Matter:** As far as modern observations show, dark matter does not interact with light or electromagnetism, so it passes right through the Earth without collision. To an electromagnetic sensor, the Earth is a solid wall; to dark matter, that wall does not exist. Yet it remains coupled through gravity: the Earth's mass bends its path.
- **Neutrinos:** A solar neutrino passes through vast distances of solid lead with almost no chance of hitting an atom. The lead is neither a solid wall nor pure nothingness; it is an area of extremely weak coupling.

C. Classical Buddhist Accounts

This spectrum of relational coupling makes sense of traditional Buddhist accounts without needing a hidden substrate:

- **Spirits Passing Through Mountains:** Spirits whose subtle bodies have no coarse physical density pass effortlessly through solid stone. Solidity is not an intrinsic property of rock; it is a resistance actualized only when two systems that share the relevant kind of coupling meet.
- **The Formless Realms:** Beings in deep formless meditative absorption have turned off coarse sensory organs; physical objects simply do not function relative to their cognitive frame.

The bowl of liquid remains a practical reference point. It is not erased just because an uncoupled observer fails to see it. But what appears there—a solid barrier, refreshing drink, burning pus, or open space—depends entirely on the channels enacted in the meeting.

D. Across Realms: The Water Offering

The hardest test of this view is interaction between different worlds, which Buddhist practice takes for granted. In traditional water offerings to hungry ghosts, water is blessed with a mantra and offered, and tradition holds that it reaches them as soothing nectar. If each realm arises within its own karmic perception, how can an action in one realm reach another?

Not by a physical substance traveling between worlds. Nothing travels from the human bowl into the hungry ghost's mouth, just as the flame at dusk does not physically travel into the flame at midnight. The giving is designated in the human world (water, a mantra, a generous intention); the relief is designated in the hungry ghost's world (nectar instead of pus). The two are connected through dependent arising—as a condition in one frame and a result in another. In one respect this resembles cross-perspective links: connection across viewpoints without any hidden substance beneath. (Part II, §3B, explains where that resemblance ends.)

7. The Myth of the Container

Why do we cling so stubbornly to the idea of a hidden object? The root obstacle is the "Container Myth"—the classical intuition of absolute space and time:

1. We imagine space as an empty, ready-made three-dimensional box.
2. We imagine placing a solid object in the center of the box.
3. We place observers inside the box and assume they must either see the real object or suffer an illusion.

Both modern physics and the Middle Way dismantle this box completely:

- **Modern Physics:** In Einstein's General Relativity, spacetime is not an empty box; it is the gravitational field itself—a physical fabric that interacts with matter and energy. In loop quantum gravity, one leading research program, space itself is pictured as a network of relations.
- **The Middle Way:** In Chapter 5 of the *Mūlamadhyamakakārikā*, Nāgārjuna shows that space cannot exist prior to its defining characteristics (such as the absence of obstruction). Space and the objects within it are designated together.

When space ceases to be an absolute warehouse, the puzzle of how an object can appear differently across species completely dissolves. Space, time, observer, and observed are all designated together, and none of them is the container of the rest.

8. Why Is There Order at All?

This brings us to the final question realism asks: "If there is no ultimate, objective ground, why is there an orderly, reliable web of cause and effect at all? Why don't things dissolve into complete chaos?"

A. The Everyday Answer: Dependent Regularity

The Middle Way does not leave order unexplained. Regularity is accounted for by the lawful nature of dependent origination:

- Causes produce matching effects: barley seeds grow barley, not rice; fire burns and does not freeze.
- Among living beings, shared karma and shared biology shape how appearances unfold.

Everyday reality has its own robust validity. Being empty of independent existence does not make it chaotic or lawless. Emptiness does not cancel cause and effect, nor is emptiness an engine that makes cause and effect work. Things never needed an intrinsic essence in order to function, and something that had an unchangeable essence of its own could never change, grow, or act.

B. Dismantling the Question

When pressed further—"Yes, but why is there an entire web of causes at all?"—the question makes a subtle mistake. It treats the entire web of dependent arising as if it were a single, bounded thing requiring an outside cause.

1. **"The Whole Web" Is a Concept:** A totality does not exist as an independent entity standing alongside its parts. Asking for an outside ground for "everything" is seeking an impossible foundation outside of all relationships.
2. **No First Beginning:** If you invent an ultimate first cause to explain the web, that cause must either depend on a prior cause (continuing the chain endlessly) or have no causes at all (in which case, having no relationships, it could never produce anything). Nāgārjuna shows that dependent arising is beginningless. An endless series is a problem only if you are anxiously seeking a final foundation. Once you stop demanding an absolute floor, having no first beginning is simply what dependent arising looks like.

Conclusion: Working Without a Foundation

Recognizing that similar beings see a similar world because they share similar conditions—and not because of a mind-independent physical bedrock—is neither nihilism nor private fantasy. It does not collapse into idealism, because the mind is just as dependent and empty as the objects it perceives. It does not collapse into lawlessness, because everyday truth functions with checkable regularity along shared channels of contact. What is eliminated is solely the anxious demand for an absolute foundation.

This insight carries profound consequences for compassion. If the hungry ghost's pus is genuinely pus for the hungry ghost, its suffering is not a mere optical illusion to be dismissed from our human vantage point. No single perspective owns the "real world," so no being's lived experience can be brushed aside. This is why compassion makes complete sense with-

out a fixed self: other beings' worlds are just as real for them as ours is for us, and emptiness takes nothing away from their suffering or from our wish to relieve it.

As Nāgārjuna declared in verse 24.14 of his Middle Way treatise: *"For whom emptiness works, everything works."* And in living experience, when appearances are recognized as unobstructed display without an inside subject grasping an outside object, reality is met as it always was: vivid, unproduced, and never requiring an absolute foundation.

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PART II: A MULTI-DIMENSIONAL WORLDVIEW

Many Worlds Without a Container

Introduction: What "Many Worlds" Means Here

Part I ended with a single bowl of liquid that is water for a human, pus for a hungry ghost, an atmosphere for a fish, nectar for a god, and Māmakī for an awakened being. If we take that seriously, a larger picture follows. Each kind of being does not merely see one object differently; it lives in a whole world of its own: its own textures, distances, dangers, and nourishment. Reality, on this view, is not one world viewed through many filters. It is many worlds, none of which is the hidden original of the others.

This Part sketches that multi-dimensional worldview. Before starting, three things need to be said plainly.

What the words mean. In this Part, a *world* is the whole field of appearances that opens up when a particular kind of being meets its surroundings through particular senses, habits, and channels of contact. Buddhist texts call this a *loka*; the biologist Jakob von Uexküll called it an *Umwelt*. When we say "multi-dimensional," we use "dimension" in its everyday sense, a whole side or aspect of reality. We do not mean the dimensions of physics (length, breadth, height, time, or the extra dimensions of string theory). Where there is any risk of confusion, this Part says "world."

What kind of claim this is. This is a conventional description, built from the reasoning of Part I. It is not a physical theory, it makes no new predictions, and nothing in modern physics confirms it. Where physics and biology appear, they appear as illustrations of a pattern, as they did in Part I.

What it does not assume. The classical Buddhist examples (hungry ghosts, gods, formless realms) belong to a traditional cosmology that many readers will not accept. The argument does not depend on them. Everything essential can be followed using animals alone: the tick, the bat, the bee, and the human. Readers who accept the traditional realms can extend the

same reasoning to them; readers who do not can stop at the animals and lose nothing structural.

The Part asks four questions in turn. What is a world? Where are the other worlds? How do worlds touch? And does any world have the final say?

1. What a World Is

A. Worlds Are Made by Channels, Not by Walls

Uexküll's tick, met in Part I, responds to three signals: the smell of butyric acid in mammal sweat, the warmth of a body, and the touch of hair. Those three signals are, almost entirely, its world. Other animals offer further examples:

- **The bat** finds its way by echolocation. Its world is shaped by returning sound: textures, edges, and the flutter of moths, all in darkness that is no obstacle to it.
- **The honeybee** sees ultraviolet light. Flowers that look plain to us carry bright landing patterns for the bee, and its world is organized by them.
- **The human** lives in a world of color, mid-sized objects, spoken words, clocks, and maps.

None of these worlds is separated from the others by a wall or a distance. They differ because each kind of being is coupled to its surroundings through different channels. What a being can sense, what it can do, and what it habitually reaches for together decide what shows up for it as real.

Buddhist thought says the same thing in its own terms. Classical Buddhist cosmology describes six realms of rebirth (gods, demigods, humans, animals, hungry ghosts, and hell beings). Their differences are explained above all by karma: the accumulated habits of intention and action that shape how a being meets everything around it.

B. Three Sides of a World, Designated Together

Every world has three sides:

1. **The being**, with its senses, body, and habits.
2. **The environment as it is met**: water to drink, pus to recoil from, a scent trail to follow.
3. **The channels of contact** that join the two.

Following Part I, none of these comes first. The being is not a finished observer who later meets an environment; the environment is not a finished stage awaiting observers; the channels are not pipes laid down in advance. All three are designated together, in dependence on one another.

Classical Abhidharma has a pair of terms for this: the *sattva-loka*, the "world of beings," and the *bhājana-loka*, the "vessel world," the environment that beings with shared karma inhabit together. The word "vessel" can sound like the very container that Part I rejected. But in the

Buddhist account, the vessel world is itself shaped by the shared karma of the beings who live in it. It is a shared environment, not an absolute box that would exist with nobody in it.

C. Worlds Come in Degrees

Worlds are not sealed compartments. They overlap to the extent that channels are shared.

- A human and a dog share sight, hearing, and touch to a large degree, and smell to a small degree (the dog's world of scent is far richer). Their worlds overlap heavily, which is why we can live, play, and work together.
- A human and a tick share very little. We meet briefly at the skin, and almost nothing else of our worlds crosses over.
- In the traditional account, a human and a hungry ghost share almost no coarse channels, which is why the same bowl can be water to one and pus to the other.

So the question "Are these the same world or different worlds?" has no single yes-or-no answer. It is a matter of how much is shared, along which channels, and for what purpose. This is exactly what Part I called *degrees of contact*.

2. Where Are the Other Worlds?

A. The Tempting Answer and Why It Fails

Once we accept that many worlds exist, a question arises at once: *where* are they? The tempting answer is: "Right here, in the same space as ours. The preta world and the human world occupy exactly the same location."

That answer sounds open-minded, but it quietly brings back the very thing Part I took apart. To say that two worlds occupy "the same location" is to assume a single, neutral space underneath both of them, a coordinate grid that exists by itself and that both worlds are placed inside. That neutral grid is the Container Myth of Part I, §7, in a new form.

Where would the "same location" be measured from? If we measure with human rulers, the answer is a human answer. To the tick, "here" is not a point on a map; it is a rising gradient of butyric acid. Each world has its own "here," its own near and far. There is no neutral position from which to lay one world's "here" on top of another's and declare that they coincide.

B. "Not Elsewhere" Is Not the Same as "In the Same Place"

The Middle Way offers a precise way out. Indian and Tibetan logic distinguish two kinds of negation:

- An **implicative negation** denies one thing and implies another. "He is not a monk" suggests he is a layperson.
- A **non-implicative negation** (Sanskrit *prasajya-pratiṣedha*, Tibetan *med dgag*) simply denies, and implies nothing in its place. Tsongkhapa held that emptiness itself is a

negation of this kind: it removes inherent existence and puts nothing else in its place.

The multi-dimensional view should use the second kind. We can say: *the other worlds are not elsewhere.* They are not in a distant place, not on another planet, not behind a hidden wall. That rejects the idea of distance. But we do not then need to add, "so they are in the same place." That second statement would bring back a shared container. The honest position is to deny the distance and leave it there.

In practice, of course, we speak from the human world, and from here it is perfectly sensible to say, "The hungry ghost is right beside you." That is a correct use of everyday language within the human frame. The error is only in thinking it describes how things are from no frame at all.

C. Different Traditions, Different Maps

Buddhist traditions themselves describe the realms in two quite different ways, and it helps to see both.

- **The Abhidharma map.** Vasubandhu's *Abhidharmakośa* (the *Treasury of Abhidharma*) lays out a vertical cosmos: Mount Meru at the center, continents around it, the god realms above, and the hell realms below, with each realm of rebirth, including that of the hungry ghosts, given its own place. This is a spatial map.
- **The co-presence map.** Huayan, Tiantai, Vajrayāna, and Dzogchen sources (discussed in Part III) stress instead that the realms, and even the pure lands, are not far away. What separates them is the way of seeing, not distance.

Neither map needs to be treated as the true layout of an absolute space. The Abhidharma map organizes meditation, ethics, and the stages of rebirth; the co-presence map points to how quickly experience can change when perception changes. Read in the light of Part I, both are conventional maps, each useful for its purpose. Neither is the container that holds everything else.

3. How Worlds Touch

If worlds differ by their channels, can they affect one another at all? Yes, in three ways, and it is important to be clear about which parts of the account rest on observation and which rest on tradition.

A. Through Shared Channels

Where channels are shared, worlds touch directly. The tick reaches the human through skin and warmth. A dog and its owner meet through sight, sound, and touch. In physics, as Part I described, dark matter passes through the Earth without striking it, yet still feels its gravity: uncoupled along one channel, coupled along another.

This kind of contact is ordinary and observable. It needs no special doctrine.

B. Through Conditions Without a Pipeline

The traditional water offering raises a harder case. In Buddhist practice, water blessed with a mantra and offered with generous intention is said to reach hungry ghosts as soothing nectar. Here the claimed connection does *not* run through any shared physical channel. Nothing travels from the human bowl into the hungry ghost's mouth.

How can such a connection be understood? The Middle Way account, as in Part I, §6D, is that the giving is designated in the human world and the relief in the hungry ghost's world, and the two are linked as condition and result within dependent arising. The *Questions of King Milinda* uses the image of a lamp burning through the night: the flame in the last watch is neither the same flame as in the first watch nor a different one, yet the one depends on the other. Dependent arising allows a condition in one frame and a result in another, without a substance passing between them.

Two honest qualifications are needed here.

- **This rests on tradition, not on observation.** That the offering actually relieves a hungry ghost is a claim of Buddhist practice and testimony. Nothing in this framework demonstrates it. What the framework shows is only that such a connection is *not ruled out* by the absence of a hidden substance. If anything, a hidden substance would make it harder to explain.
- **The physics parallel is limited.** Part I compared cross-realm connection to the "cross-perspective links" of relational quantum mechanics. That comparison holds in one respect only: in both, agreement or connection between perspectives does not require a hidden, perspective-free substrate. But in Adlam and Rovelli's postulate, observers come to agree by *physically interacting* with each other. The water offering is described as having no such physical channel. So the parallel is structural, not evidential: relational quantum mechanics does not support the efficacy of offerings, and should not be cited as if it did.

C. What Does Not Cross

One thing does not cross between worlds: a view from nowhere. No being can step outside every world and compare them all from a neutral overview, because any comparison is itself made from somewhere, through some channel. A human can learn a great deal about the bat's world through instruments and experiments, but always as a human, building a human picture of the bat. This is not a failing to be fixed. It is simply what it means for every view to be a view from somewhere.

4. Does Any World Have the Final Say?

A. No Master World Among Ordinary Worlds

Ordinary realism tempts us to believe that the human world, with its colors, solid objects, and three spatial dimensions, is the "base reality," and that other beings register reduced or distorted versions of it. Part I has already removed the ground for that belief. If there is no hidden original, the human world cannot be the original either. It is one world among many: coherent and reliable within its own channels, but not the master copy of the others.

This calls for humility, not dismissal of the human view. The human world is the one we live in, think in, and act in, and for us it carries every practical priority. What it lacks is *ultimate* priority: the right to declare the bat's world, the bee's world, or the hungry ghost's world mere errors measured against ours.

B. Science Extends the Human World; It Does Not Leave It

A natural objection is that science escapes the human frame. After all, it tells us about ultraviolet light the bee can see and we cannot, and about neutrinos that no animal senses at all.

Science does something remarkable: it *extends* human channels. Spectrometers, microscopes, and particle detectors couple us to things our unaided senses miss. In this sense science is the most reliable map we possess within its domains. But every instrument is built, read, and interpreted by humans, and every result is expressed in human concepts. Science widens the human world enormously, but it does not lift us out of every world into a view from nowhere. This is fully consistent with Part I's statement that scientific models are "practical designations, among the most reliable we possess."

C. Not Everything Is Equal

"No master world" does not mean that every perception is as good as every other. Part I drew two distinctions, and both still hold.

1. **Within a world, there are errors.** A human with jaundice who sees a white conch as yellow is mistaken by human standards. A bee with damaged eyes is mistaken by bee standards. Each world has its own norms of healthy and impaired perception, and errors are corrected within it.
2. **Between ordinary and awakened vision, there is a real difference.** In Mipham's account (Part I, §4D), pure vision is not just one more ordinary world competing with the others. It is the way *any* world appears when the habit of grasping at solid subjects and objects has dropped away: where how things appear matches how they actually are. It ranks above ordinary perception not because it reaches a hidden foundation, but because it no longer distorts.

So the multi-dimensional view is not a flat relativism in which anything goes. It is horizontal among ordinary worlds, and it recognizes a real difference between confusion and clarity.

5. Comparison With Everett's Many-Worlds Interpretation

"Many worlds" also names a well-known interpretation of quantum mechanics. Because the phrases are the same, it is worth setting out carefully how the two ideas differ and where they meet.

A. A Fair Description of Everett

In 1957 Hugh Everett III proposed what he called the "relative state" formulation of quantum mechanics, now known as the Many-Worlds Interpretation (MWI). Its central claims, as most of its present-day defenders state them, are:

- **The wavefunction never collapses.** A single universal quantum state evolves smoothly and *deterministically* according to the Schrödinger equation. Many-Worlds does not remove determinism; it is one of the most deterministic interpretations available. What it removes is the idea that only one outcome happens.
- **Branches emerge through decoherence.** When a quantum system becomes entangled with its environment, the universal state effectively divides into branches that no longer interfere with one another in any practical way. Each branch contains a version of the observer seeing one outcome. On the view defended by David Wallace and others, these branches are *emergent and approximate*, not fundamental objects.
- **Branches are not separate places.** Branches are not stacked side by side in physical space. They are components of a single quantum state. The mathematical "space" in which that state is described, Hilbert space, is an abstract space of possible states, not a place where things are located.

Everettians also disagree among themselves about how literally to take the wavefunction's reality, how probabilities arise, and what exactly a branch is. The description above is a fair outline, not a settled consensus.

B. Where the Two Views Meet

Both views reject the common-sense picture of a single world seen in a single way. Both say that the world of any one observer is not the whole story. And on Wallace's reading, Everett's worlds, like the Buddhist worlds of this Part, are not fundamental building blocks but patterns that emerge from something more basic. These are real resonances.

C. Where They Differ

The deepest difference is not about physics at all, but about what is taken to be ultimately real.

- **The floor.** Most defenders of Many-Worlds are realists about the universal wavefunction: it is the one thing that really exists, from which the branches emerge. The Middle Way accepts no such floor. The multi-dimensional view of this Part stands on nothing beneath the worlds, not even a mathematical one.

- **How worlds arise.** Everett's branches multiply over time as quantum events decohere. The worlds of this Part do not split from one another; they arise together with the beings whose channels enact them.
- **Contact between worlds.** Once decohered, Everett's branches practically never influence one another again. The worlds of this Part touch along shared channels, and in the traditional account, through dependent conditions such as offerings.

A caution belongs here too. This comparison concerns the *interpretation* of quantum mechanics, not its working equations, which every interpretation shares. Physics as practiced does not decide between realism about the wavefunction and the Middle Way. If a physical interpretation is to be set beside the Middle Way, the closer companion is not Everett but Rovelli's relational quantum mechanics, which has one world of relative facts rather than many branches, and which Rovelli himself has compared to Nāgārjuna.

D. Summary Table

Aspect	Everett's Many-Worlds (as usually defended)	Buddhist multi-dimensional view (this Part)
What is ultimately real	Usually, the universal wavefunction	Nothing found under analysis; worlds are conventionally real
Determinism	Fully deterministic evolution of the universal state	Dependent arising: regular, but neither strict determinism nor chance
What a "world" is	An emergent, approximate branch of one quantum state	The field of appearances enacted by a kind of being through its channels
Where worlds are	Not in physical space; components of one state described in abstract Hilbert space	Not elsewhere, and not "in the same place" either; location is designated within each world
How worlds arise	Branches multiply through decoherence over time	Worlds arise together with the beings who live in them
Contact between worlds	Practically none after decoherence	Through shared channels; in tradition, also through dependent conditions
Status of the comparison	A physical interpretation, debated among physicists	A philosophical and contemplative description, not a physical theory

6. The Limits of This View

Part I was careful to say what it did not prove. This Part should be equally clear.

1. **It does not prove that other realms exist.** It shows how such realms *could* be understood without a hidden container, if they exist. Belief in hungry ghosts, gods, and hell realms rests on Buddhist tradition and contemplative testimony, not on this argument. The animal examples stand without them.

2. **Physics does not confirm it.** Uexküll, dark matter, neutrinos, and relational quantum mechanics illustrate patterns: channels, degrees of coupling, and agreement without a substrate. They are not evidence for Buddhist cosmology, and most of the scientists who developed them would not endorse that use.
3. **It is not a scientific theory.** It makes no measurable predictions and offers no mechanism. It is a way of describing reality that removes a confusion: the idea of one hidden original.
4. **It is not relativism.** Errors are still errors within each world, and the difference between confusion and awakened clarity remains.
5. **It is not idealism.** Worlds are not inside minds. Minds, environments, and channels arise together, and none of them is the foundation.
6. **It is one reading within Buddhism.** Buddhist schools disagree on these points. Tibetan commentators such as Tsongkhapa and Gorampa read Candrakīrti's bowl example differently, and the Abhidharma, Yogācāra, Madhyamaka, and Dzogchen traditions each describe the realms in their own way. What is presented here follows the Middle Way as read through Tsongkhapa and Mipham, as in Part I.

7. Living in a Many-World Universe

What difference does this view make, beyond philosophy?

Humility. If the human world is one world among many, then what we see is not the measure of all that is. Every creature we meet, from a bee to a stranger, lives in a world we cannot fully enter.

Compassion. As Part I concluded, if the hungry ghost's pus is genuinely pus for the hungry ghost, its suffering is real within its world and cannot be dismissed from ours. The same holds, without any appeal to tradition, for the fear of an animal or the loneliness of a person whose world we do not share. The many-world view does not weaken compassion; it removes one of its excuses: "That isn't really happening."

Practice. In the Vajrayāna and Dzogchen traditions discussed in Part III, the shift from ordinary to pure vision is not a journey to another place. Nothing needs to be moved; what changes is the way of meeting what is already here. That is exactly what the multi-dimensional view would lead us to expect: if worlds differ by their channels and not by distance, then transformation is a change in seeing, not a change in location.

Freedom from the floor. Finally, this view carries the conclusion of Part I to its widest scale. Not only the single bowl, but the whole range of worlds, needs no hidden foundation beneath it. The worlds hold together through dependent arising, and through nothing else. As Nāgārjuna said, *"For whom emptiness works, everything works."*

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PART III: CLASSICAL BUDDHIST TRADITIONS ON MANY WORLDS

Abhidharma, Yogācāra, Huayan, Tiantai, Vajrayāna, and Dzogchen

Introduction: How to Read These Traditions

Part II sketched a multi-dimensional worldview using the Middle Way as read by Tsongkhapa and Mipham. Buddhism, however, has thought about many worlds for more than two thousand years, in India, China, and Tibet. This Part surveys six of those traditions.

Two cautions apply throughout.

They do not all say the same thing. It is tempting to present every Buddhist school as a witness for one shared view. That would be inaccurate. Some of these traditions hold positions that the Middle Way explicitly criticizes; others use language that a careful Madhyamaka reader would want to qualify. Each section below therefore asks three questions: What does this tradition say about many worlds? Where does it agree with the view of Parts I and II? And where does it differ, or need care?

They are not physics. These traditions describe worlds in terms of karma, perception, meditation, and awakening. They are not early versions of modern cosmology, and they did not anticipate modern physics. Where they resemble Part II, the resemblance is in the shape of the reasoning, not in any scientific content.

The traditions are taken roughly in historical order.

1. Abhidharma: The Map of the Realms

A. What It Says

The Abhidharma ("higher teaching") traditions organized the Buddha's teachings into systematic lists and maps. The most influential summary is Vasubandhu's *Abhidharmakośa* (the *Treasury of Abhidharma*), written from the standpoint of the Sarvāstivāda and Sautrāntika schools, before Vasubandhu's turn to Yogācāra.

Its third chapter lays out the cosmos in detail:

- **Three realms:** the desire realm (which contains hell beings, hungry ghosts, animals, humans, and the lower gods), the form realm of subtle meditative gods, and the formless realm of pure meditative absorption.
- **A vertical geography:** Mount Meru at the center, four continents around it, the god realms rising above, and the hells lying beneath. Each realm of rebirth, including that of the hungry ghosts, is given its own place on this map.

- **Two kinds of world:** the *sattva-loka* (the world of beings) and the *bhājana-loka* (the "vessel world," their shared environment).

Crucially, the fourth chapter opens with the statement that the variety of the world is born of action (*karmajaṃ lokavaicitryam*). The vessel world is not a pre-existing box into which beings are placed. It is formed by the shared karma of the beings who will live in it, and it decays when that karma is exhausted.

B. Where It Agrees

The Abhidharma already holds the key idea of Part II: worlds are shaped by the actions and habits of the beings who inhabit them. Even the "vessel" is a product of shared karma, not an independent container.

C. Where It Differs

The Abhidharma map is strongly spatial. It describes realms stacked above and below one another, separated by measured distances. Its philosophy also differs from the Middle Way: the Sarvāstivāda school held that the ultimate building blocks of experience, the *dharmas*, have their own intrinsic nature (*svabhāva*). That is precisely the belief Nāgārjuna set out to refute.

Read in the light of Part II, the Abhidharma map is best taken as a conventional map that serves meditation, ethics, and the understanding of rebirth. It is not a description of an absolute space.

2. Yogācāra: Worlds as Shared Ripening

A. What It Says

The Yogācāra ("practice of yoga") school, founded by Asaṅga and Vasubandhu in the fourth and fifth centuries, explains shared worlds through consciousness.

- **Shared perception without an external object.** As Part I described, Vasubandhu's *Twenty Verses* (*Viṃśatikā*) uses the hungry ghosts who all see the same river of pus to show that shared perception does not require an external object. Beings with similar karma experience similar worlds.
- **The storehouse consciousness.** Yogācāra explains this through the *ālaya-vijñāna*, a deep layer of consciousness that carries karmic seeds from life to life. Asaṅga's *Compendium of the Mahāyāna* (*Mahāyānasaṃgraha*) distinguishes "common" seeds, which ripen as the shared environment, from "uncommon" seeds, which ripen as each being's own body and senses.
- **One situation, many seeings.** Later Yogācāra writers return again and again to the hungry ghosts and the river, using them to show that beings with different karma meet the same situation in different ways: an early form of the bowl example.

B. Where It Agrees

Yogācāra gives the most detailed classical account of *why* beings of one kind share a world. Its idea of common and uncommon seeds explains both the overlap and the difference between worlds, much as Part II's account of shared and unshared channels does.

C. Where It Differs

Yogācāra grounds these shared worlds in the storehouse consciousness. As Part I, §2D explained, Candrakīrti rejected this move: if the external object is not established by itself, neither is the mind that perceives it. Making consciousness the foundation is, from the Middle Way perspective, simply another floor.

Two qualifications are fair to Yogācāra. First, some modern scholars read Yogācāra less as a metaphysical idealism ("only mind exists") and more as an analysis of how experience is constructed. Second, later Indian thinkers, notably Śāntarakṣita, combined Yogācāra and Madhyamaka, accepting Yogācāra conventionally while denying that consciousness is ultimately real. On either reading, Yogācāra is best treated here as a valuable partner in the conversation, and on the decisive question of a foundation, as a contrast rather than an ally.

3. Huayan: The Non-Obstruction of Phenomena

A. What It Says

The Huayan ("Flower Ornament") school of China takes its name from the *Avataṃsaka Sūtra*, a vast scripture filled with images of worlds within worlds. In the sūtra, buddha-fields as numerous as atoms appear within a single dust mote, and no world crowds out another.

Huayan thinkers developed these images into philosophy:

- **Dushun** (557–640), traditionally counted as the first patriarch, is credited with the *Contemplation of the Dharmadhātu*, which leads the practitioner from emptiness, through the non-obstruction of principle and phenomena, to the all-pervading inclusion of everything in everything.
- **Fazang** (643–712), the school's great systematizer, explained mutual inclusion through images such as the golden lion and the hall of mirrors, and through the image of **Indra's Net**: a net with a jewel at every knot, in which each jewel reflects all the others, and none is the central or original jewel.
- **Chengguan** (738–839) set out the teaching of the **Four Dharmadhātus**, the four ways of seeing reality:
    1. **Shi (事):** the realm of distinct, everyday phenomena.
    2. **Li (理):** the realm of principle, or emptiness.
    3. **Li-shi wu'ai (理事無礙):** the non-obstruction of principle and phenomena: form is emptiness, and emptiness is form.
    4. **Shi-shi wu'ai (事事無礙):** the non-obstruction of phenomena with one another.

The fourth view is the key one for many worlds. Because no phenomenon has an independent self-nature, phenomena do not exclude one another. Each depends on all the others, and so, in Huayan's language, each contains all the others.

B. Where It Agrees

Huayan is the clearest classical statement that worlds do not obstruct one another *because* they are empty. That is exactly the reasoning of Part II: worlds that have no solid, independent core have nothing with which to push each other away. Indra's Net, in which no jewel is the original, is a fine image for Part II's "no master world."

C. Where It Needs Care

Two points need care.

- **Non-obstruction is not a claim about space.** The "dust mote containing infinite realms" is not a statement that countless worlds are packed into one tiny location of a shared space. It expresses mutual dependence: each thing is what it is only in relation to everything else. Reading it spatially would bring back the container of Part I, §7.
- **The status of "principle."** Huayan also teaches that phenomena arise from the one true nature (*xingqi*, 性起), closely linked to buddha-nature. Some readers, ancient and modern, have taken this "principle" as a positive, all-embracing ground. A Middle Way reading insists that principle is emptiness, which is not a ground of anything. Huayan can be read either way, and this essay adopts the second reading.

4. Tiantai: The Mutual Inclusion of the Ten Realms

A. What It Says

The Tiantai school, systematized by **Zhiyi** (538–597), builds on Nāgārjuna directly. Its teaching of the **Three Truths** (emptiness, the provisional, and the middle) develops a famous verse of the *Mūlamadhyamakakārikā* (24.18), which says that dependent arising is emptiness, is a dependent designation, and is the middle way.

Tiantai's teaching on worlds has two parts:

- **The mutual inclusion of the ten realms** (*shijie huju*, 十界互具). There are ten realms of existence: hell beings, hungry ghosts, animals, demigods (*asuras*), humans, and gods (the six realms of rebirth), together with *śrāvakas*, *pratyekabuddhas*, bodhisattvas, and buddhas. Each realm contains all ten. A human contains the potential of hell and the potential of buddhahood; a hell being is not cut off from awakening.
- **Three thousand realms in a single thought-moment** (*yinian sanqian*, 一念三千). In Zhiyi's *Great Calming and Contemplation* (*Mohe zhiguan*), the ten realms, each containing ten, multiplied by further factors, give three thousand aspects of existence, all present in a single moment of mind.

B. Where It Agrees

Tiantai makes many worlds immediate and personal. The realms are not remote places; they are possibilities present in each moment of experience. Which one becomes vivid depends on habit, intention, and practice. This matches Part II's conclusion that transformation is a change in seeing, not a change in location.

C. Where It Needs Care

- **A thought-moment is not a place.** The "three thousand realms in a single thought-moment" concern a moment of experience, not a point in space. They should not be read as saying that a hell and a pure land occupy the same spatial location.
- **"Inherent" inclusion.** Later Tiantai, especially in the Song dynasty, emphasized that the Buddha *inherently includes* (*xingju*, 性具) even the hell realms and evil. This bold teaching is distinctive to Tiantai and is stronger than anything the Indian Middle Way asserts. It is valuable as a remedy for treating buddhahood as a distant, separate realm, but it is Tiantai's own position, not a general Buddhist one.

5. Vajrayāna: Pure Vision and the Mandala

A. What It Says

In the Vajrayāna traditions of India and Tibet, the theme of many worlds becomes a method of practice. Its root is already present in the Mahāyāna sūtras. In the first chapter of the *Vimalakīrti Sūtra*, Śāriputra wonders why the Buddha's land looks so impure, full of stones, thorns, and filth. The Buddha touches the ground with his toe, and the whole world appears as a jewelled pure land. The Buddha explains that the land was always pure; Śāriputra had not seen it. As the sūtra says, when the mind is pure, the buddha-field is pure.

Vajrayāna develops this into the practice of **pure vision** (Tibetan *dag snang*):

- In meditation, the practitioner trains in seeing the environment as the mandala palace of the deity, beings as deities, sounds as mantra, and thoughts as wisdom.
- The elements are recognized as wisdom energies, often personified as female buddhas. In the Nyingma tradition, the water element is Māmakī, as in the bowl example of Part I.
- This is understood not as pretending, but as training to see what the Vimalakīrti story says was always there.

B. Where It Agrees

Pure vision is Part II's "change in seeing, not in location" put into practice. Nothing needs to be moved, rebuilt, or traveled to. What changes is the way the same field of experience is met, as in Mipham's account of pure conventional valid knowing (Part I, §4D).

C. Where It Needs Care

- **Not all pure lands are described as "here."** Many Mahāyāna sources describe pure lands as real places in other directions; Amitābha's Sukhāvatī, for example, is said to lie countless buddha-fields to the west. Vajrayāna and the *Vimalakīrti Sūtra* emphasize that purity is a matter of perception, but they do not simply deny the traditional descriptions. Both ways of speaking coexist within Buddhism.
- **This essay describes a view, not a method.** In Vajrayāna, pure vision is cultivated within a lineage, with empowerment and a qualified teacher. Nothing here is meant as practice instruction.

6. Dzogchen: Unimpeded Openness

A. What It Says

In the Dzogchen (Great Perfection) teachings of Longchenpa (1308–1364) and Ju Mipham (1846–1912), described in Part I, §4, the basis is empty in essence (*ka dag*, primordial purity) and spontaneously present in nature (*lhun grub*). Appearances are the expressive power or dynamism (*rtsal*) of the basis, not products manufactured by it.

Dzogchen texts characterize this openness as *zang thal*: unimpeded, transparent, all-penetrating. Because no appearance has a solid core, appearances do not block one another. The many worlds of beings can all arise without crowding anything, because the basis is not an extended physical space that could fill up. It is not a space at all.

B. Where It Agrees

Dzogchen, as read in Part I, is the closest of these traditions to the view of this essay. Its insistence that the basis is a "ground" only conventionally, that awareness is not a projector, and that display is not causal production, protects the many-worlds view from sliding into either a cosmic container or a cosmic mind.

C. Where It Needs Care

Dzogchen has long been controversial within Tibetan Buddhism. Critics have argued that its language of a "basis" or "ground" reintroduces a foundation, or resembles non-Buddhist doctrines of an eternal self. Longchenpa and Mipham answered these charges by insisting that the basis is empty and is called a ground only conventionally, which is the reading followed here. Readers should know, however, that this reading is defended, not simply assumed, within the tradition itself.

7. Reading the Traditions Together

The six traditions can be summarized as follows.

Tradition	Key idea about many worlds	Where it agrees with this essay	Where it differs or needs care
Abhidharma	Realms formed by the shared karma of their inhabitants (*karmajaṃ lokavaicitryam*)	The "vessel world" is shaped by beings, not a pre-existing box	Strongly spatial map; holds that *dharmas* have their own intrinsic nature
Yogācāra	Shared worlds as the ripening of common karmic seeds	Detailed account of why beings share or do not share a world	Grounds worlds in the storehouse consciousness, which Candrakīrti rejected
Huayan	Non-obstruction of all phenomena; Indra's Net	Worlds do not obstruct one another because they are empty	Not a spatial claim; "principle" must not become a positive ground
Tiantai	Ten realms mutually included; three thousand realms in one thought-moment	Worlds are possibilities of experience, not distant places	A thought-moment is not a location; "inherent inclusion" is Tiantai's own bold claim
Vajrayāna	Pure vision; the world as mandala	Transformation is a change in seeing, not in location	Traditional pure lands also described as distant; a practice requiring a teacher
Dzogchen	Unimpeded openness (*zang thal*) of empty, spontaneously present appearances	Closest to this essay; no container and no cosmic mind	The "basis" is read as a ground only conventionally, a reading the tradition itself has had to defend

What do these traditions share, despite their differences? Three things stand out.

1. **Worlds are shaped by beings.** In every tradition surveyed, the realms are shaped by the karma, perception, or realization of those who live in them. None treats the world as a ready-made stage that exists the same way for everyone.
2. **Worlds need not obstruct one another.** From Huayan's Indra's Net to Dzogchen's unimpeded openness, the absence of a solid core is what lets many worlds arise without crowding.
3. **Transformation is a change in seeing.** From the *Vimalakīrti Sūtra* to Tiantai and pure vision, liberation is presented not as travel to a better place, but as meeting this same field of experience without distortion.

What they do not share is an answer to the question of a foundation. Abhidharma accepts intrinsically real *dharmas*; Yogācāra accepts a foundational consciousness; Huayan and Dzogchen can be read with or without a positive ground. The view of this essay chooses the reading without a foundation, following Nāgārjuna, Candrakīrti, Tsongkhapa, and Mipham. That is a choice made for reasons, given in Part I, not a consensus the traditions already share.

The traditions, then, are not proofs of the multi-dimensional worldview, any more than physics was. They are a long conversation, carried on across many centuries and cultures,

about the same puzzle with which this essay began: one bowl, many beings, and no hidden original beneath them.

SOURCES AND FURTHER READING

Texts Referred to in This Article

Buddhist Sources

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- **Tsongkhapa**, *Illumination of the Thought* (*dGongs pa rab gsal*), his commentary on *Entering the Middle Way*.
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- **Asaṅga**, *Mahāyānasaṃgraha* (*Compendium of the Mahāyāna*).
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- **Carlo Rovelli**, "Relational Quantum Mechanics," *International Journal of Theoretical Physics* (1996); *Helgoland* (2021).

- **Emily Adlam and Carlo Rovelli**, “Information Is Physical: Cross-Perspective Links in Relational Quantum Mechanics,” *Philosophy of Physics* (2023).
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